27881번 - Gahui and ILGAM lake 다국어

시간 제한	메모리 제한	제출	정답	맞힌 사람	정답 비율
1 초	1024 MB	99	39	37	51.389%

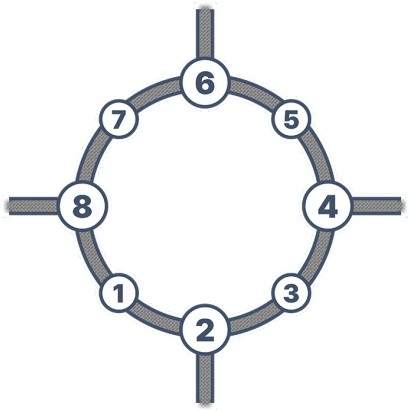
문제

Gahui often visited Konkuk University. Every time Gahui visited Konkuk University, she was surprised to see the huge `ILGAM` lake. It is bigger than several universities, such as Seokyung university and Hanseong university.

The lake has $4n$ points arranged like a ring, each numbered from 1 to $4n$ in `counter-clockwise` order. Each pair of adjacent points are connected in both directions. In other words:

- If $1 \le a < 4n$, point a is connected to point $a+1$ in **both directions.**
- Point 1 is connected to point $4n$ in **both directions.**

Points n, $2n$, $3n$, and $4n$ are connected to 3 subway stations (`Konkuk_Univ` , `Guui` , `Sejong_Univ`) via roads. `ILGAM` lake is so big, the nearest station is different depending on the point. Gahui wants to know which subway station is the nearest to her current point. Please help Gahui.



[Picture 1] Roads and points when n = 2

입력

n is given in the first line.

In the next line, $4n$ numbers are given in the following format:

$$d_1 \quad d_2 \quad \dots \quad d_{4n}$$

- d_i is distance from point i to point $i+1$ when $1 \le i < 4n$
- d_{4n} is distance from point $4n$ to point 1

And the information below is given on four lines.

$$d_{n,\texttt{Konkuk_Univ}} \quad d_{n,\texttt{Guui}} \quad d_{n,\texttt{Sejong_Univ}}$$

$$d_{2n,\texttt{Konkuk_Univ}} \quad d_{2n,\texttt{Guui}} \quad d_{2n,\texttt{Sejong_Univ}}$$

$$d_{3n,\texttt{Konkuk_Univ}} \quad d_{3n,\texttt{Guui}} \quad d_{3n,\texttt{Sejong_Univ}}$$

$$d_{4n,\texttt{Konkuk_Univ}} \quad d_{4n,\texttt{Guui}} \quad d_{4n,\texttt{Sejong_Univ}}$$

$d_{a,b}$ is the **`distance between point a and station b.`**

In the next line, `Q` (number of questions) is given.

In the next `Q` lines, `Q` questions are given in the following format, one per line.

- `k`
  - What is **the distance to the nearest station** from point `k`?

출력

Print the answers to the questions, **one per line.**

제한

- $1 \le$ **all distances** in the problem $\le 10^5$
- $1 \le n \le 10^5$
- $1 \le Q \le 10^5$

예제 입력 1 복사

```
1
1 1 1 1
999 999 999
888 888 888
1 2 3
888 888 888
1
1
```

예제 출력 1 복사

```
3
```

Move from point `1` to point `3`. Then, move from point `3` to `Konkuk_Univ` Station.

예제 입력 2 복사

```
1
100 90 80 1
999 999 999
888 888 888
777 777 777
1 1 1
1
1
```

예제 출력 2 복사

```
2
```

출처

Contest (/category/45) > BOJ User Contest (/category/984) > 가희와 함께 하는 코딩 테스트 (/category/529) > 가희와 함께 하는 5회 코딩테스트 (/category/detail/3565) E번

- 문제를 만든 사람: chogahui05 (/user/chogahui05)
- 문제를 검수한 사람: chansol (/user/chansol), eaststar (/user/eaststar), egod1537 (/user/egod1537), kiwiyou (/user/kiwiyou), utilforever (/user/utilforever)